Food-Delivery-SQL-Analytics-Queries

CREATE DATABASE ZOMATO;

USE ZOMATO;

SHOW TABLES FROM ZOMATO;

SELECT * FROM CUSTOMERS;

DESCRIBE CUSTOMERS;

ALTER TABLE CUSTOMERS

MODIFY COLUMN CUSTOMER_ID INT PRIMARY KEY,

MODIFY COLUMN NAME VARCHAR(35),

MODIFY COLUMN EMAIL VARCHAR(50),

MODIFY COLUMN PHONE VARCHAR(15),

MODIFY COLUMN LOCATION VARCHAR(500),

MODIFY COLUMN SIGNUP_DATE DATE,

MODIFY COLUMN PREFERRED_CUISINE VARCHAR(20),

MODIFY COLUMN TOTAL_ORDERS INT,

MODIFY COLUMN AVERAGE_RATING DECIMAL(4,2);

SELECT * FROM DELIVERIES;

DESCRIBE DELIVERIES;

ALTER TABLE DELIVERIES

MODIFY COLUMN DELIVERY_ID INT PRIMARY KEY,

MODIFY COLUMN ORDER_ID INT,

MODIFY COLUMN DELIVERY_PERSON_ID INT,

MODIFY COLUMN DELIVERY_STATUS VARCHAR(30),

MODIFY COLUMN DISTANCE DECIMAL(5,2),

MODIFY COLUMN DELIVERY_TIME INT,

MODIFY COLUMN ESTIMATED_TIME INT,

MODIFY COLUMN DELIVERY_FEE DECIMAL(7,2),

MODIFY COLUMN VEHICLE_TYPE VARCHAR(30);

SELECT * FROM DELIVERY_PERSONS;

ALTER TABLE DELIVERY_PERSONS

MODIFY COLUMN DELIVERY_PERSON_ID INT PRIMARY KEY,

MODIFY COLUMN NAME VARCHAR(50),

MODIFY COLUMN CONTACT_NUMBER VARCHAR(20),

MODIFY COLUMN VEHICLE_TYPE VARCHAR(20),

MODIFY COLUMN TOTAL_DELIVERIES INT,

MODIFY COLUMN AVERAGE_RATING DECIMAL(5,2),

MODIFY COLUMN LOCATION VARCHAR(100);

SELECT * FROM RESTAURANTS;

ALTER TABLE RESTAURANTS

MODIFY COLUMN RESTAURANT_ID INT PRIMARY KEY,

MODIFY COLUMN NAME VARCHAR(50),

MODIFY COLUMN CUISINE_TYPE VARCHAR(50),

MODIFY COLUMN LOCATION VARCHAR(100),

MODIFY COLUMN OWNER_NAME VARCHAR(50),

MODIFY COLUMN AVERAGE_DELIVERY_TIME INT,

MODIFY COLUMN CONTACT_NUMBER VARCHAR(15),

MODIFY COLUMN RATING DECIMAL(5,2),

MODIFY COLUMN TOTAL_ORDERS INT,

MODIFY COLUMN IS_ACTIVE INT;

```
SELECT * FROM ORDERS;
ALTER TABLE ORDERS
ADD COLUMN ORDER_DATED DATE,
ADD COLUMN ORDER_TIME TIME;

SET SQL_SAFE_UPDATES = 0;
UPDATE ORDERS
SET
    ORDER_DATED = DATE(ORDER_DATE),
    ORDER_TIME = TIME(ORDER_DATE);

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM ORDERS;

ALTER TABLE ORDERS
MODIFY COLUMN ORDER_DATED DATE AFTER RESTAURANT_ID;

ALTER TABLE ORDERS
MODIFY COLUMN ORDER_TIME TIME AFTER ORDER_DATED;
ALTER TABLE ORDERS
DROP COLUMN ORDER_DATE;
SELECT * FROM ORDERS;
ALTER TABLE ORDERS
RENAME COLUMN ORDER_DATED TO ORDER_DATE;
```

SELECT * FROM ORDERS;

ALTER TABLE ORDERS

ADD COLUMN DELIVERY_DATE DATE,

ADD COLUMN DELIVERY_TIMES TIME;

SET SQL_SAFE_UPDATES = 0;

UPDATE ORDERS

SET

        DELIVERY_DATE = DATE(DELIVERY_TIME),

  DELIVERY_TIMES = TIME(DELIVERY_TIME);

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM ORDERS;

SET SQL_SAFE_UPDATES = 1;

ALTER TABLE ORDERS

DROP COLUMN DELIVERY_TIME;

SELECT * FROM ORDERS;

ALTER TABLE ORDERS

MODIFY COLUMN DELIVERY_DATE DATE AFTER ORDER_TIME;

ALTER TABLE ORDERS

MODIFY COLUMN DELIVERY_TIMES TIME AFTER DELIVERY_DATE;

```sql
SELECT * FROM ORDERS;

DESCRIBE ORDERS;

ALTER TABLE ORDERS
MODIFY COLUMN ORDER_ID INT PRIMARY KEY,
MODIFY COLUMN CUSTOMER_ID INT,
MODIFY COLUMN RESTAURANT_ID INT,
MODIFY COLUMN STATUS VARCHAR(20),
MODIFY COLUMN TOTAL_AMOUNT DECIMAL(10,2),
MODIFY COLUMN PAYMENT_MODE VARCHAR(20),
MODIFY COLUMN DISCOUNT_APPLIED DECIMAL(5,2),
MODIFY COLUMN FEEDBACK_RATING DECIMAL(5,2);

ALTER TABLE ORDERS
ADD CONSTRAINT FK_CUSTOMER_ID
FOREIGN KEY (CUSTOMER_ID)
REFERENCES CUSTOMERS(CUSTOMER_ID);

ALTER TABLE ORDERS
ADD CONSTRAINT FK_RESTAURANT_ID
FOREIGN KEY (RESTAURANT_ID)
REFERENCES RESTAURANTS(RESTAURANT_ID);

DESCRIBE ORDERS;
```

```
DESCRIBE DELIVERIES;

ALTER TABLE DELIVERIES
ADD CONSTRAINT FK_ORDER_ID
FOREIGN KEY (ORDER_ID)
REFERENCES ORDERS(ORDER_ID);

ALTER TABLE DELIVERIES
ADD CONSTRAINT FK_DELIVERY_PERSON_ID
FOREIGN KEY (DELIVERY_PERSON_ID)
REFERENCES DELIVERY_PERSONS(DELIVERY_PERSON_ID);

DESCRIBE RESTAURANTS;

describe orders;

SELECT * FROM ORDERS;

SELECT STATUS,COUNT(STATUS) AS COUNT FROM ORDERS
GROUP BY STATUS;
```

-- 1. Which customers generate the highest revenue?

```
SELECT      C.CUSTOMER_ID AS CUSTOMER_ID,
            C.NAME AS CUSTOMER_NAME ,
            ROUND(SUM(O.TOTAL_AMOUNT),2) AS REVENUE_GENERATED,
            round(AVG(O.TOTAL_AMOUNT),2) AS AVERAGE_AMOUNT,
            COUNT(O.ORDER_ID) AS ORDER_COUNT
FROM ORDERS O
JOIN CUSTOMERS C
ON C.CUSTOMER_ID = O.CUSTOMER_ID
WHERE O.STATUS = "DELIVERED"
GROUP BY C.CUSTOMER_ID,C.NAME
ORDER BY REVENUE_GENERATED DESC
LIMIT 5;
```

CUSTOMER_ID	CUSTOMER_NAME	REVENUE_GENERATED	AVERAGE_AMOUNT	ORDER_COUNT
436	Dhriti Bhatnagar	13845.66	2769.13	5
333	Wishi Sani	13471.06	3367.77	4
463	Hardik Bansal	13351.34	3337.84	4
227	Hitesh Pathak	13241.54	2648.31	5
127	Victor Mitra	12146.06	4048.69	3

-- Customer 436 generated the highest revenue of ₹13,845.66 from 5 delivered orders, with an average order value of ₹2,769.13.

```
-- 2. Who are the top 10 customers by number of orders?

SELECT   C.CUSTOMER_ID AS CUSTOMER_ID,
         C.NAME AS CUSTOMER_NAME,
         COUNT(ORDER_ID) AS ORDERS_COUNT
FROM ORDERS O
JOIN CUSTOMERS C
ON O.CUSTOMER_ID=C.CUSTOMER_ID
WHERE O.STATUS = "DELIVERED"
GROUP BY C.CUSTOMER_ID,C.NAME
ORDER BY ORDERS_COUNT DESC
LIMIT 10;
```

CUSTOMER_ID	CUSTOMER_NAME	ORDERS_COUNT
227	Hitesh Pathak	5
92	Anjali Dara	5
436	Dhriti Bhatnagar	5
333	Wishi Sani	4
98	Eshana Chander	4

-- The most active customers were Customer IDs 227,92 AND 436.

-- Also customer_id 436 appeared in top 1 highest revenue generate list with frequently orders

```
-- 3.Which customers have high order frequency but low spending?

SELECT  C.CUSTOMER_ID AS CUSTOMER_ID,
        C.NAME AS CUSTOMER_NAME,
        COUNT(O.ORDER_ID) AS ORDER_COUNT,
   ROUND(SUM(O.TOTAL_AMOUNT),2) AS AMOUNT
FROM CUSTOMERS C
```

JOIN ORDERS O

ON C.CUSTOMER_ID = O.CUSTOMER_ID

WHERE O.STATUS = 'DELIVERED'

GROUP BY C.CUSTOMER_ID,C.NAME

ORDER BY ORDER_COUNT DESC,AMOUNT ASC

LIMIT 5;

CUSTOMER_ID	CUSTOMER_NAME	ORDER_COUNT	AMOUNT
92	Anjali Dara	5	6007.39
227	Hitesh Pathak	5	13241.54
436	Dhriti Bhatnagar	5	13845.66
476	Ayaan Malhotra	4	7299.16
215	Sudiksha Sha	4	8292.36

-- Customer_id 92 have high frequency in ordering but customer is low spending.We can attract her with more campaigns

-- Cutsomer_id 436 stand outs again with high frequency of orders with high revenue along side with customer_id 227

-- 4. Which premium customers generate the most revenue?

SELECT * FROM CUSTOMERS;

SELECT C.CUSTOMER_ID AS CUSTOMER_ID,

C.NAME AS CUSTOMER_NAME,

COUNT(O.ORDER_ID) AS COUNT_ORDERS,

SUM(O.TOTAL_AMOUNT) AS REVENUE_GENERATED

FROM CUSTOMERS C

JOIN ORDERS O

ON C.CUSTOMER_ID = O.CUSTOMER_ID

WHERE O.STATUS = 'DELIVERED'

AND   C.IS_PREMIUM = 1

GROUP BY C.CUSTOMER_ID,C.NAME

ORDER BY REVENUE_GENERATED DESC

LIMIT 5;

CUSTOMER_ID	CUSTOMER_NAME	COUNT_ORDERS	REVENUE_GENERATED
436	Dhriti Bhatnagar	5	13845.66
34	Zayyan Madan	4	12028.65
459	Yahvi Sane	3	11964.96
303	Meera Sidhu	3	11566.00
263	Harinakshi Kapoor	3	10257.30

-- Customer_id 436 being our regular and high revenue generating customer among all the customers.

SELECT MAX(ORDER_DATE) FROM ORDERS;

-- 5.Identify customers whose order frequency has declined significantly over time.

```
WITH CUSTOMERS_BEHAVIOUR AS(
                    SELECT  C.CUSTOMER_ID AS CUSTOMER_ID,
                                    C.NAME AS CUSTOMER_NAME,
            COUNT(CASE WHEN O.ORDER_DATE >= DATE('2025-07-12') - interval 1 YEAR
                THEN 1 END) AS RECENT_ORDERS,
 COUNT(CASE WHEN O.ORDER_DATE < DATE('2025-07-12') - interval 1 YEAR
                                                THEN 1 END ) AS OLD_ORDERS
                                    FROM ORDERS O JOIN CUSTOMERS C
                                    ON O.CUSTOMER_ID = C.CUSTOMER_ID
            WHERE O.STATUS = "DELIVERED"
            GROUP BY C.CUSTOMER_ID,C.NAME
)
SELECT * FROM CUSTOMERS_BEHAVIOUR
WHERE OLD_ORDERS > RECENT_ORDERS
```

```sql
ORDER BY (OLD_ORDERS -  RECENT_ORDERS) DESC;
```

CUSTOMER_ID	CUSTOMER_NAME	RECENT_ORDERS	OLD_ORDERS
464	Omya Sant	0	3
303	Meera Sidhu	0	3
266	Ekaraj Suri	0	3
223	Ekavir Chada	0	3
114	Hredhaan Vyas	0	2
226	Bhavika Wason	0	2
476	Ayaan Malhotra	1	3
294	Akshay Karan	0	2
319	Jagdish Dutta	0	2
170	Vanya Goda	0	2
222	Daniel Majumdar	0	2

-- Customer_id 464,303,266,223 didnot order any orders in last 1 year

-- **6. What percentage of total revenue comes from premium customers versus regular customers?**

```sql
SELECT
   C.IS_PREMIUM,
   ROUND(
      SUM(O.TOTAL_AMOUNT) * 100 /
      (SELECT SUM(TOTAL_AMOUNT) FROM ORDERS),
      2
   ) AS REVENUE_PERCENTAGE
FROM CUSTOMERS C
JOIN ORDERS O
   ON C.CUSTOMER_ID = O.CUSTOMER_ID
WHERE O.STATUS = 'DELIVERED'
GROUP BY C.IS_PREMIUM;
```

IS_PREMIUM	REVENUE_PERCENTAGE
0	16.93
1	15.99

-- 7. Which restaurants generate the highest revenue despite having average ratings?(TOP 3)

```
SELECT  R.RESTAURANT_ID AS RESTAURANT_ID,
                    R.NAME AS RESTAURANT_NAME,
                    ROUND(SUM(O.TOTAL_AMOUNT),2) AS AMOUNT_GENERATED,
      ROUND(SUM(O.TOTAL_AMOUNT)/R.TOTAL_ORDERS,2) AS AVG_AMOUNT_PER_ORDER,
                    R.CUISINE_TYPE AS CUISINE_TYPE,
                    R.AVERAGE_DELIVERY_TIME AS AVG_DELIVERY_TIME,
                    R.TOTAL_ORDERS AS TOTAL_ORDERS,
      R.RATING AS RATING
FROM ORDERS O JOIN RESTAURANTS R
ON R.RESTAURANT_ID=O.RESTAURANT_ID
WHERE R.IS_ACTIVE = 1 AND
           R.RATING BETWEEN 2.00 AND 3.50 AND
     O.STATUS = 'DELIVERED'
GROUP BY R.RESTAURANT_ID,R.NAME
ORDER BY AMOUNT_GENERATED DESC;
```

RESTAURANT_ID	RESTAURANT_NAME	AMOUNT_GENERATED	AVG_AMOUNT_PER_ORDER	CUISINE_TYPE	AVG_DELIVERY_TIME	TOTAL_ORDERS	RATING
106	Classic Cafe (Desserts)	23346.43	115.58	Continental	67	202	2.72
36	Flavors of Spot (Chinese)	20247.95	12.77	Chinese	64	1585	2.67
23	Golden Corner (South Indian)	16825.76	59.04	South Indian	100	285	2.41
118	Flavors of Bistro (Mediterranean)	15134.59	5.75	Continental	56	2634	2.42
125	Madras House (Thai)	14161.66	23.64	Chinese	109	599	2.77
38	Chennai Kitchen (Mexican)	13937.59	21.61	Chinese	49	645	2.58
77	Chennai Corner (Bakery)	12609.49	5.01	North Indian	61	2518	2.59
28	Chennai Kitchen (French)	11219.33	10.18	Italian	65	1102	2.18
124	Spicy Diner (Continental)	10891.52	11.21	South Indian	7	972	3.04
12	Tandoori Lounge (Desserts)	9667.11	4.16	North Indian	90	2326	2.78
198	Delicious Diner (French)	9210.46	24.30	Italian	66	379	2.21

-- AND R.AVERAGE_DELIVERY_TIME < 100

-- Out of 28 restaurants, 24 restaurants are having less avg_delivery_time(100).This indicates people are preferring more for fast deliveries than good rating.

-- 8.Which restaurants have excellent ratings but surprisingly low order volumes?

```
SELECT  O.RESTAURANT_ID AS RESTAURANT_ID,
                    R.NAME AS RESTAURANT_NAME,
      SUM(O.TOTAL_AMOUNT) AS AMOUNT_GENERATED,
      COUNT(O.RESTAURANT_ID) AS TOTAL_ORDERS,
      R.RATING AS RATING,
      ROUND(SUM(O.TOTAL_AMOUNT)/COUNT(O.RESTAURANT_ID),2) AS
AMOUNT_PER_ORDER,
      R.AVERAGE_DELIVERY_TIME AS AVG_DELIVERY_TIME
FROM RESTAURANTS R
 JOIN ORDERS O
ON R.RESTAURANT_ID = O.RESTAURANT_ID
WHERE   R.RATING > 4.25
          AND O.STATUS = 'DELIVERED'
GROUP BY O.RESTAURANT_ID,R.NAME
ORDER BY TOTAL_ORDERS ASC;
```

RESTAURANT_ID	RESTAURANT_NAME	AMOUNT_GENERATED	TOTAL_ORDERS	RATING	AMOUNT_PER_ORDER	AVG_DELIVERY_TIME
11	Classic Kitchen (Street Food)	2900.21	1	4.40	2900.21	17
13	Royal House (Juice Bar)	378.63	1	4.94	378.63	49
78	Tandoori Hut (Italian)	2633.04	1	4.46	2633.04	26
88	Authentic Kitchen (BBQ)	4453.16	1	4.32	4453.16	17
100	Delicious Grill (Desserts)	4297.41	1	4.73	4297.41	31
101	Royal Lounge (BBQ)	488.77	1	4.53	488.77	102
151	Authentic Corner (Biryani)	759.13	1	4.98	759.13	28
10	Flavors of Bistro (Chinese)	6162.43	2	4.27	3081.22	34
19	Grand Eatery (Italian)	7759.01	2	4.84	3879.51	89
21	Classic Cafe (Mexican)	4103.10	2	4.95	2051.55	83
57	Golden Grill (Mediterranean)	3532.88	2	4.44	1766.44	86

Result 12

-- These are the Restaurants generating less revenue desspite having excellent rating

-- Amount_per_order and Avg_delivery_time can be reasons for less revenue.

-- 9. Rank restaurants within each cuisine category based on revenue generated.

```
SELECT * FROM RESTAURANTS;

WITH CUISINE AS (
                SELECT  R.RESTAURANT_ID AS RESTAURANT_ID,
                R.NAME AS RESTAURANT_NAME,
    R.CUISINE_TYPE AS CUISINE_TYPE,
                ROUND(SUM(O.TOTAL_AMOUNT),2) AS REVENUE_GENERATED,
                RANK() OVER(PARTITION BY CUISINE_TYPE ORDER BY SUM(O.TOTAL_AMOUNT)
DESC)RA
FROM RESTAURANTS R JOIN ORDERS O ON R.RESTAURANT_ID = O.RESTAURANT_ID
WHERE O.STATUS = 'DELIVERED'
                AND R.IS_ACTIVE = 1
GROUP BY R.RESTAURANT_ID,R.NAME,R.CUISINE_TYPE)
                SELECT RESTAURANT_ID,
    RESTAURANT_NAME,
    CUISINE_TYPE,RA,REVENUE_GENERATED
FROM CUISINE
WHERE RA =1;
```

RESTAURANT_ID	RESTAURANT_NAME	CUISINE_TYPE	RA	REVENUE_GENERATED
36	Flavors of Spot (Chinese)	Chinese	1	20247.95
106	Classic Cafe (Desserts)	Continental	1	23346.43
142	Chennai Grill (Biryani)	Italian	1	14450.67
16	Chennai Grill (Street Food)	North Indian	1	16766.53
97	Spicy Bistro (South Indian)	South Indian	1	17470.76

-- These are Restuarants Tops 1 in their respective Cuisine_type.

-- 10. Which restaurants receive the highest customer ratings while maintaining fast delivery times?

```
SELECT  R.RESTAURANT_ID AS RESTAURANT_ID,R.NAME AS RESTAURANT_NAME,
                ROUND(AVG(D.DELIVERY_TIME),2) AS AVG_DELIVERY_TIME_PER_ORDER,
    ROUND(AVG(FEEDBACK_RATING),2) AS AVG_FEEDBACK_RATING
    FROM RESTAURANTS R JOIN ORDERS O ON R.RESTAURANT_ID = O.RESTAURANT_ID
    JOIN DELIVERIES D ON O.ORDER_ID = D.ORDER_ID
    WHERE D.DELIVERY_STATUS = 'DELIVERED'
    AND R.IS_ACTIVE = 1
    GROUP BY R.RESTAURANT_ID,R.NAME
    HAVING AVG_DELIVERY_TIME_PER_ORDER < 30.00
    AND AVG_FEEDBACK_RATING > 4.00
                ORDER BY AVG_DELIVERY_TIME_PER_ORDER ASC,
                            AVG_FEEDBACK_RATING DESC;
```

RESTAURANT_ID	RESTAURANT_NAME	AVG_DELIVERY_TIME_PER_ORDER	AVG_FEEDBACK_RATING
159	Savor Eatery (French)	15.00	4.48
42	Royal Eatery (Cafe)	18.67	4.31
97	Spicy Bistro (South Indian)	21.00	4.33
176	Authentic Corner (Desserts)	23.00	4.42
195	Classic House (Mediterranean)	23.00	4.32

-- These restaurants maintain fast delivery times while achieving high customer ratings.

-- Savor Eatery (French) and Royal Eatery (Cafe) show the best balance between delivery speed and customer satisfaction.

-- 11. What percentage of total platform revenue is contributed by the top 5 restaurants?

```
WITH TOP_5 AS(
SELECT  O.RESTAURANT_ID,ROUND(SUM(O.TOTAL_AMOUNT),2) AS TOTAL_5
    FROM ORDERS O JOIN RESTAURANTS R ON R.RESTAURANT_ID = O.RESTAURANT_ID
    WHERE O.STATUS = 'DELIVERED'
             AND   R.IS_ACTIVE = 1
    GROUP BY O.RESTAURANT_ID)
SELECT ROUND(SUM(TOTAL_5)/(SELECT SUM(TOTAL_5) FROM TOP_5),2)*100 AS PERCENTAGE
FROM (SELECT * FROM TOP_5
ORDER BY TOTAL_5 DESC
LIMIT 5) T;
```

	PERCENTAGE
▶	14.00

-- Top 5 Restaurants are generating 14% of revenue from overall revenue

-- 12. Which delivery partners consistently complete deliveries faster than estimated time?

```
WITH CTE AS (SELECT  D.DELIVERY_PERSON_ID AS DELIVERY_PERSON_ID,
             DP.NAME AS DELIVERY_PERSON_NAME,
    COUNT(*) AS TOTAL_DELIVERIES,
    SUM(CASE WHEN DELIVERY_TIME < ESTIMATED_TIME THEN 1
             ELSE 0 END) AS COUNT
    FROM DELIVERIES D JOIN DELIVERY_PERSONS DP ON DP.DELIVERY_PERSON_ID =
D.DELIVERY_PERSON_ID
             WHERE DELIVERY_STATUS = 'DELIVERED'
    GROUP BY D.DELIVERY_PERSON_ID,DP.NAME
    HAVING COUNT(*) > 2
```

```
    ORDER BY COUNT DESC,TOTAL_DELIVERIES DESC)
SELECT  DELIVERY_PERSON_ID,DELIVERY_PERSON_NAME,
            TOTAL_DELIVERIES,COUNT,
            ROUND((COUNT/TOTAL_DELIVERIES)*100,2) AS FASTER_DELIVERY_PERCENTAGE
    FROM CTE
    GROUP BY DELIVERY_PERSON_ID,DELIVERY_PERSON_NAME
    ORDER BY FASTER_DELIVERY_PERCENTAGE DESC,
                        TOTAL_DELIVERIES DESC
            LIMIT 5;
```

DELIVERY_PERSON_ID	DELIVERY_PERSON_NAME	TOTAL_DELIVERIES	COUNT	FASTER_DELIVERY_PERCENTAGE
125	Ryan Soman	3	3	100.00
273	Darpan Bava	3	3	100.00
157	Hemang Guha	3	3	100.00
432	Naksh Khurana	3	3	100.00
123	Chasmum Gole	4	3	75.00

-- 13. Which delivery partners handle the highest workload while maintaining high ratings?

```
SELECT DELIVERY_PERSON_ID,NAME,TOTAL_DELIVERIES,AVERAGE_RATING FROM
DELIVERY_PERSONS
WHERE AVERAGE_RATING >= 4.00
GROUP BY DELIVERY_PERSON_ID,NAME
ORDER BY TOTAL_DELIVERIES DESC,
            AVERAGE_RATING DESC
    LIMIT 10;
```

DELIVERY_PERSON_ID	NAME	TOTAL_DELIVERIES	AVERAGE_RATING
107	Gabriel Sharaf	298	4.14
492	Nilima Kamdar	297	4.00
71	Thomas Dubey	291	4.62
272	Yutika Mitter	291	4.13
28	Oliver Mall	290	4.33
240	Faris Ravel	286	4.45
308	Amaira Tak	285	4.43
102	Kritika Edwin	285	4.39
44	Andrew Gara	285	4.15
325	Mohini Dar	279	4.96
NULL	NULL	NULL	NULL

-- 14. Which vehicle type provides the most efficient deliveries based on distance and time?

```
WITH VEHICLE AS (SELECT  VEHICLE_TYPE,
                COUNT(DELIVERY_ID) AS VEHICLES_COUNT,
                SUM(DISTANCE) AS TOTAL_DISTANCE,
    SUM(DELIVERY_TIME) AS TOTAL_TIME_IN_MINS
FROM DELIVERIES
WHERE DELIVERY_STATUS = 'DELIVERED'
GROUP BY VEHICLE_TYPE)
                SELECT  VEHICLE_TYPE,VEHICLES_COUNT,
                ROUND((TOTAL_DISTANCE/TOTAL_TIME_IN_MINS)*1000,2) AS
METERS_COVERED_BY_VEHICLE_IN_1_MIN
                FROM VEHICLE
    GROUP BY VEHICLE_TYPE
    ORDER BY METERS_COVERED_BY_VEHICLE_IN_1_MIN DESC;
```

VEHICLE_TYPE	VEHICLES_COUNT	METERS_COVERED_BY_VEHICLE_IN_1_MIN
Electric Bike	75	787.37
Scooter	71	731.25
Bike	44	683.17
Moped	72	638.11
Electric Scooter	69	634.11
Motorcycle	59	591.52

-- Electric Bikes demonstrate the highest delivery efficiency, covering 787 meters per minute on average

-- 15. What percentage of deliveries are completed within the estimated delivery time?

```
SELECT * FROM DELIVERIES;

SELECT ROUND(SUM(CASE WHEN DELIVERY_TIME < ESTIMATED_TIME THEN 1
                ELSE 0 END) * 100.0/ COUNT(*),2) AS FAST_DELIVERY_PERCENTAGE
                FROM DELIVERIES
                WHERE DELIVERY_STATUS = 'DELIVERED';
```

	FAST_DELIVERY_PERCENTAGE
▶	64.36

-- Fast delivery percentage is 64.36 which was impressive

-- 16. How does monthly revenue trend over time and what months show significant growth or decline?

```
WITH MONTHLY_REVENUE AS(
SELECT  MONTH(ORDER_DATE) AS MONTH_NUMBER,
                MONTHNAME(ORDER_DATE) AS MONTH_NAME,
    ROUND(SUM(TOTAL_AMOUNT),2) AS REVENUE
FROM ORDERS
WHERE STATUS = 'DELIVERED'
GROUP BY MONTH_NUMBER,MONTH_NAME
)
SELECT  MONTH_NUMBER,MONTH_NAME,REVENUE,
                LAG(REVENUE)OVER(ORDER BY MONTH_NUMBER) AS PREVIOUS_NUMBER,
    ROUND((REVENUE - LAG(REVENUE)OVER(ORDER BY MONTH_NUMBER)),2) AS DIFFERENCE,
    ROUND((REVENUE-LAG(REVENUE) OVER(ORDER BY MONTH_NUMBER))
    /(LAG(REVENUE)OVER(ORDER BY MONTH_NUMBER))*100,2) AS GROWTH_PERECENTAGE
FROM MONTHLY_REVENUE;
```

	MONTH_NUMBER	MONTH_NAME	REVENUE	PREVIOUS_NUMBER	DIFFERENCE	GROWTH_PERECENTAGE
▶	1	January	122675.36	NULL	NULL	NULL
	2	February	83184.07	122675.36	-39491.29	-32.19
	3	March	94681.02	83184.07	11496.95	13.82
	4	April	88084.46	94681.02	-6596.56	-6.97
	5	May	105680.46	88084.46	17596.00	19.98
	6	June	71528.65	105680.46	-34151.81	-32.32
	7	July	122377.55	71528.65	50848.90	71.09
	8	August	124445.38	122377.55	2067.83	1.69
	9	September	92660.16	124445.38	-31785.22	-25.54
	10	October	102903.19	92660.16	10243.03	11.05
	11	November	98157.49	102903.19	-4745.70	-4.61

-- July month major growth in revenue apporx.71% increase in the revenue

-- Wherever february and June got approx. 30% dip in the revenee

-- January,July and August have most of the revenue

-- 17. How much revenue is lost because of cancelled or unsuccessful orders?

```
SELECT  STATUS,COUNT(ORDER_ID) AS ORDER_COUNT,
                    ROUND(SUM(TOTAL_AMOUNT),2) AS REVENUE
FROM ORDERS
WHERE STATUS <> 'DELIVERED'
GROUP BY STATUS
ORDER BY REVENUE DESC ;
```

Result Grid | Filter Rows:

	STATUS	ORDER_COUNT	REVENUE
▶	Cancelled	508	1235687.08
	Pending	506	1210825.42

-- 18 .What is the impact of discounts on customer spending behavior?

```
SELECT  CASE WHEN DISCOUNT_APPLIED BETWEEN 0 AND 10 THEN '0-10'
                              WHEN DISCOUNT_APPLIED BETWEEN 11 AND 20 THEN '11-20'
                              WHEN DISCOUNT_APPLIED BETWEEN 21 AND 30 THEN '21-30'
```

```
WHEN DISCOUNT_APPLIED BETWEEN 31 AND 40 THEN '31-40'
ELSE '41-50'
END AS DISCOUNT_BUCKET,
COUNT(*) AS TOTAL_ORDERS,
ROUND(AVG(TOTAL_AMOUNT)/COUNT(*),2) AS AVG_ORDER_VALUE
FROM ORDERS
WHERE STATUS = 'DELIVERED'
GROUP BY DISCOUNT_BUCKET
ORDER BY DISCOUNT_BUCKET;
```

DISCOUNT_BUCKET	TOTAL_ORDERS	AVG_ORDER_VALUE
0-10	79	31.12
11-20	76	29.96
21-30	90	28.59
31-40	92	28.51
41-50	149	16.23

```
-- 19. Which payment methods are associated with the highest average order values?
SELECT * FROM ORDERS;
SELECT PAYMENT_MODE,COUNT(*) AS ORDER_COUNT,
ROUND(SUM(TOTAL_AMOUNT),2) AS ORDER_AMOUNT,
ROUND(SUM(TOTAL_AMOUNT)/COUNT(*),2) AS AVERAGE_ORDER_PRICE,
ROUND(SUM(TOTAL_AMOUNT)/(SELECT SUM(TOTAL_AMOUNT) FROM ORDERS
WHERE STATUS = 'DELIVERED'),2)*100 AS PERCENTAGE  FROM ORDERS
WHERE STATUS = 'DELIVERED'
GROUP BY PAYMENT_MODE
ORDER BY PERCENTAGE DESC;
```

PAYMENT_MODE	ORDER_COUNT	ORDER_AMOUNT	AVERAGE_ORDER_PRICE	PERCENTAGE
Credit Card	160	421178.61	2632.37	35.00
Cash	173	397410.24	2297.17	33.00
UPI	153	381848.91	2495.74	32.00

-- 20. Customer Lifetime Value (CLV) Ranking

SELECT * FROM ORDERS;

WITH REVENUE AS

(SELECT C.CUSTOMER_ID AS CUSTOMER_ID,C.NAME AS CUSTOMER_NAME,

COUNT(O.ORDER_ID) AS ORDERS_COUNT,

SUM(O.TOTAL_AMOUNT) AS REVENUE

FROM CUSTOMERS C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID

WHERE O.STATUS = 'DELIVERED'

GROUP BY C.CUSTOMER_ID,C.NAME

ORDER BY REVENUE DESC)

SELECT CUSTOMER_ID,CUSTOMER_NAME,

ORDERS_COUNT,REVENUE,

DENSE_RANK()OVER(ORDER BY REVENUE DESC) AS CLV_RANK

FROM REVENUE

ORDER BY CLV_RANK ;

CUSTOMER_ID	CUSTOMER_NAME	ORDERS_COUNT	REVENUE	CLV_RANK
436	Dhriti Bhatnagar	5	13845.66	1
333	Wishi Sani	4	13471.06	2
463	Hardik Bansal	4	13351.34	3
227	Hitesh Pathak	5	13241.54	4
127	Victor Mitra	3	12146.06	5

-- 21.Identify the top 20% of customers contributing approximately 80% of total revenue (Pareto Analysis).

```
SELECT * FROM CUSTOMERS;

WITH REVENUE AS (

        SELECT  C.CUSTOMER_ID AS CUSTOMER_ID,C.NAME AS CUSTOMER_NAME,

                SUM(O.TOTAL_AMOUNT) AS TOTAL_AMOUNT,

                COUNT(O.ORDER_ID) AS ORDERS_COUNT

                FROM ORDERS O JOIN CUSTOMERS C ON
O.CUSTOMER_ID=C.CUSTOMER_ID

                WHERE O.STATUS = 'DELIVERED'

                GROUP BY  C.CUSTOMER_ID,C.NAME

                ORDER BY TOTAL_AMOUNT DESC,

                        ORDERS_COUNT DESC),

SUM AS(         SELECT  CUSTOMER_NAME,CUSTOMER_ID,TOTAL_AMOUNT,

                SUM(TOTAL_AMOUNT) OVER(ORDER BY TOTAL_AMOUNT DESC)
AS CUMMUTATIVE_SUM,

    SUM(TOTAL_AMOUNT)OVER() AS TOTAL_REVENUE

    FROM REVENUE

    GROUP BY CUSTOMER_NAME,CUSTOMER_ID

    ORDER BY TOTAL_AMOUNT DESC)

 SELECT     CUSTOMER_NAME,CUSTOMER_ID,TOTAL_AMOUNT,

                ROUND((TOTAL_AMOUNT / TOTAL_REVENUE)*100,2) AS
REVENUE_PERCENT,

    ROUND((CUMMUTATIVE_SUM / TOTAL_REVENUE)*100,2) AS
CUMMUTATIVE_PERCENT

    FROM SUM

    WHERE ROUND((CUMMUTATIVE_SUM / TOTAL_REVENUE)*100,2) <= 80.00
```

ORDER BY TOTAL_AMOUNT DESC;

CUSTOMER_NAME	CUSTOMER_ID	TOTAL_AMOUNT	REVENUE_PERCENT	CUMMUTATIVE_PERCENT
Dhriti Bhatnagar	436	13845.66	1.15	1.15
Wishi Sani	333	13471.06	1.12	2.28
Hardik Bansal	463	13351.34	1.11	3.39
Hitesh Pathak	227	13241.54	1.10	4.49
Victor Mitra	127	12146.06	1.01	5.50
Zayyan Madan	34	12028.65	1.00	6.50
Yahvi Sane	459	11964.96	1.00	7.50
Falan Chaudhry	232	11658.98	0.97	8.47
Meera Sidhu	303	11566.00	0.96	9.44

Result 5 ×